

[Aim: 100/100 in Maths]

आरंभ

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Lecture #7

Digits Wale Sawaal:

#LP: The sum of the digits of a 2-digit number is 14. The number obtained by interchanging its digits exceeds the given number by 18. Find the number. [CBSE 2024]

$$\begin{aligned} \text{Let } U &= y \\ T &= x \end{aligned} \rightarrow N = x \times 10 + y \times 1$$
$$N = 10x + y$$

ATQ

ATQ

$$x + y = 14 \quad \text{--- (I)}$$

$$N' = N + 18$$

$$10y + x = 10x + y + 18$$

$$10y - x + x - 10x = 18$$

digits interchange

$$\begin{aligned} U &= x \\ T &= y \end{aligned} \rightarrow N' = y \times 10 + x \times 1$$
$$N' = 10y + x$$

$$9y - 9x = 18$$
$$9(y - x) = 18$$

$$y - x = 2$$
$$-x + y = 2 \quad \text{--- (II)}$$

$$\begin{array}{l} \textcircled{\text{I}} \rightarrow x + y = 14 \\ \textcircled{\text{II}} \rightarrow -x + y = 2 \end{array}$$

$$\begin{array}{r} x + y = 14 \\ -x + y = 2 \\ \hline 2y = 16 \\ y = 8 \end{array}$$

put $y = 8$

$$\Downarrow$$

$$x + 8 = 14$$

$$x = 14 - 8$$

$$\boxed{x = 6}$$

Final Ans

$$\begin{aligned} N &= 10x + y \\ &= 10(6) + 8 \end{aligned}$$

$$\boxed{N = 68}$$

Q3B:

$$\begin{array}{c} \text{tens} \quad \text{unit} \\ \uparrow \quad \uparrow \\ 42 = 4 \times \underline{10} + 2 \times 1 \end{array}$$

$$\begin{array}{c} t \quad u \\ 24 = 2 \times 10 + 4 \times 1 \end{array}$$

$$\begin{array}{l} U=4 \\ T=6 \end{array} \rightarrow \begin{array}{l} N = 6 \times 10 + 4 \times 1 \\ \boxed{N = 64} \end{array}$$

digit interchange

$$\begin{array}{l} U=6 \\ T=4 \end{array} \rightarrow \begin{array}{l} N' = 4 \times 10 + 6 \times 1 \\ \boxed{N' = 46} \end{array}$$

$$\begin{array}{l} U=1 \\ T=7 \end{array} \rightarrow \begin{array}{l} N = 7 \times 10 + 1 \times 1 \\ \boxed{N = 71} \end{array}$$

$$\begin{array}{l} U=7 \\ T=1 \end{array} \rightarrow \begin{array}{l} N' = 1 \times 10 + 7 \times 1 \\ \boxed{N' = 17} \end{array}$$

$$\begin{array}{l} U=6 \\ T=9 \end{array} \rightarrow \begin{array}{l} N = 9 \times 10 + 6 \times 1 \\ \boxed{N = 96} \end{array}$$

$$\begin{array}{l} U=9 \\ T=6 \end{array} \rightarrow \begin{array}{l} N' = 6 \times 10 + 9 \times 1 \\ \boxed{N' = 69} \end{array}$$

digit interchange

$$\begin{array}{l} U=y \\ T=x \end{array} \rightarrow \begin{array}{l} N = x \times 10 + y \times 1 \\ \boxed{N = 10x + y} \end{array}$$

$$\begin{array}{l} T=y \\ U=x \end{array} \rightarrow \begin{array}{l} N' = y \times 10 + x \times 1 \\ \boxed{N' = 10y + x} \end{array}$$

#LP: A two-digit number is such that the product of its digits is 12. When 36 is added to this number, the digits interchange their place. Find the number (CBSE 2025)

let unit place digit $\Rightarrow y$
tens " " $\Rightarrow x$

$$N = x \times 10 + y \times 1$$

$$N = 10x + y$$

ATQ

$$x \cdot y = 12 \quad \text{--- (I)}$$

ATQ

$$N + 36 = N'$$

$$10x + y + 36 = 10y + x$$

$$10x - x + y - 10y = -36$$

Digits interchange

$$U = x$$

$$T = y$$

$$N' = y \times 10 + x \times 1$$

$$N' = 10y + x$$

$$9x - 9y = -36$$

$$9(x - y) = -36$$

$$x - y = -4 \quad \text{--- (II)}$$

$$\textcircled{I} \Rightarrow x \cdot y = 12 \rightarrow y = \frac{12}{x}$$

$$\textcircled{II} \Rightarrow x - y = -4$$

$\Downarrow$ put

$$\rightarrow x - \frac{12}{x} = -4$$

$$\rightarrow \frac{x^2 - 12}{x} = -4$$

$$\rightarrow x^2 - 12 = -4x$$

$$\rightarrow \boxed{x^2 + 4x - 12 = 0}$$

$$\boxed{x^2 + 6x - 2x - 12 = 0}$$

$$\begin{aligned} x(x+6) - 2(x+6) &= 0 \\ (x+6)(x-2) &= 0 \end{aligned}$$

$$x+6=0$$

$$\cancel{x = -6}$$

$$x-2=0$$

$$\boxed{x=2} \checkmark$$

$$y = \frac{12}{2} = 6$$

$$\boxed{y=6} \checkmark$$

Final Ans

$$N = 10x + y$$

$$= 10(2) + 6$$

$$\boxed{N = 26} \text{ Ans}$$

#LP: The sum of the digits of a two-digit number is 13. If the number is subtracted from the one obtained by interchanging the digits, the result is 45. What is the number?

Let unit place digit = y
tens " = x

$$N = 10x + y$$

ATQ

$$x + y = 13 \quad \text{--- (I)}$$

ATQ $N' - N = 45$

$$\Rightarrow 10y + x - (10x + y) = 45$$

$$\Rightarrow 10y + x - 10x - y = 45$$

$$9y - 9x = 45$$

digits reverse
 $U = x$
 $T = y$ } $N' = 10y + x$

$$y - x = 5$$

$$-x + y = 5 \quad \text{--- (II)}$$

$$x = 4$$
$$y = 9$$

Final Ans

$$N = 10x + y$$
$$= 10(4) + 9$$

$$N = 49$$

~~Q. 1. mp~~
#LP: The sum of a two-digit number and the number formed by reversing the order of digits is 66. If the two digits differ by 2, find the number. How many such numbers are there?

Let $U = y$
 $T = x$ } $N = 10x + y$

→ digits rev.

$U = x$
 $T = y$ } $N' = 10y + x$

Ans

$$N + N' = 66$$

$$10x + y + 10y + x = 66$$

$$11x + 11y = 66$$

$$11(x + y) = 66$$

$x + y = 6$ — (I)

Ans $x - y = 2$ — (II)

$x = 4$
 $y = 2$

$N = 10x + y$
 $= 10 \times 4 + 2$
 $N = 42$

Two no.s are possible 42 & 24

$y - x = 2$ — (II)

$x = 2$
 $y = 4$

$N = 10x + y$
 $= 10(2) + 4$
 $N = 24$

#LP: If three times the greater of two numbers is divided by the smaller one, we get 4 as the quotient and 3 as the remainder. Also, if seven times the smaller number is divided by the greater one, we get 5 as the quotient and 1 as the remainder. Find the numbers. [CBSE 2024]

Let greater no. is x ($x > y$)
smaller no. is y .

ATQ

$$y \overline{) 3x} \begin{array}{l} 4 \\ \underline{3} \end{array}$$

$$\text{Div} = \text{divisor} \times Q + R$$

$$3x = y \times 4 + 3$$

$$3x = 4y + 3$$

$$\boxed{3x - 4y = 3} \text{--- (I)}$$

ATQ

$$x \overline{) 7y} \begin{array}{l} 5 \\ \underline{5} \end{array}$$

$$D = d \times q + r$$

$$7y = x \times 5 + 1$$

$$7y = 5x + 1$$

$$\boxed{7y - 5x = 1} \text{--- (II)}$$

Final Ans

Greater $\Rightarrow x =$
Smaller $\Rightarrow y =$

#LP: Sum of two numbers is 105 and their difference is 45. Find the numbers. **[CBSE 2024]**

Let x & y ($x > y$)

$$x + y = 105$$

$$x - y = 45$$

→ एक के बाद एक

#LP: Three consecutive odd integers are such that the sum of the squares of the first two numbers is greater than the square of the third by 68. Find the numbers. [CBSE 2023]

(Example में दी सीरीज)

let $\begin{matrix} 1 & & 3 & & 5 \\ \uparrow & & \uparrow & & \uparrow \\ (x) & , & (x+2) & , & (x+4) \\ 1 & & 2 & & 3 \end{matrix}$

A7Q

$$(1)^2 + (2)^2 = (3)^2 + 68$$

$$(x)^2 + (x+2)^2 = (x+4)^2 + 68$$

$x \sim$

#LP: Half of the difference between two numbers is 2. The sum of the greater number and twice the smaller number is 13. Find the numbers. [CBSE 2023]

Let greater $\Rightarrow x$ ($x > y$)
Smaller $\Rightarrow y$

ATQ

$$\frac{1}{2}(x - y) = 2$$

$$\boxed{x - y = 4} \text{--- (I)}$$

ATQ

$$\boxed{x + 2y = 13} \text{--- (II)}$$

$$\left. \begin{array}{l} x = 7 \\ y = 3 \end{array} \right\} \checkmark$$

Revise notes

THANK YOU COODIESS !!

